

Lecture 7: Object Recognition

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Outline

- Introduction to Object Recognition
 - Visual recognition tasks
 - Challenges
- Bias-Variance Trade-off
- A Simple Object Recognition Pipeline



What are the different visual recognition tasks?



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Classification: Does this image contain a building?



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Classification: Is this a beach?

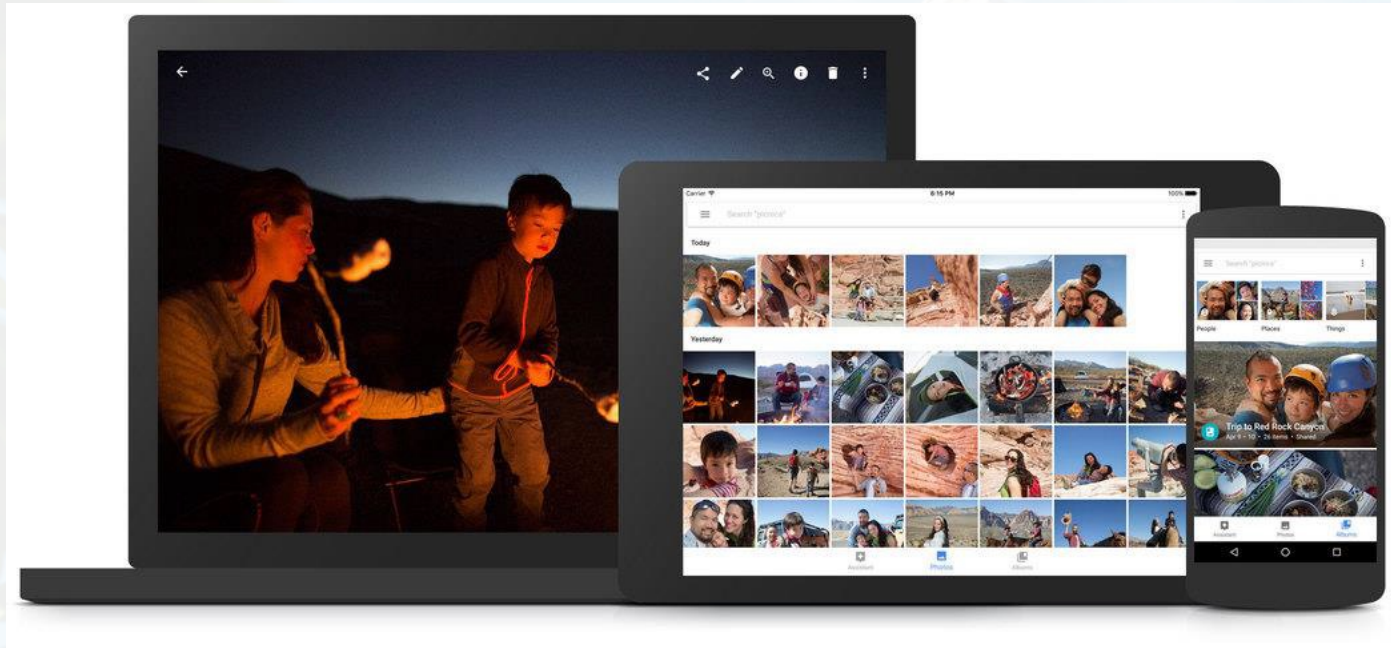


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Applications: Searching & Indexing Images

Organizing photo collections



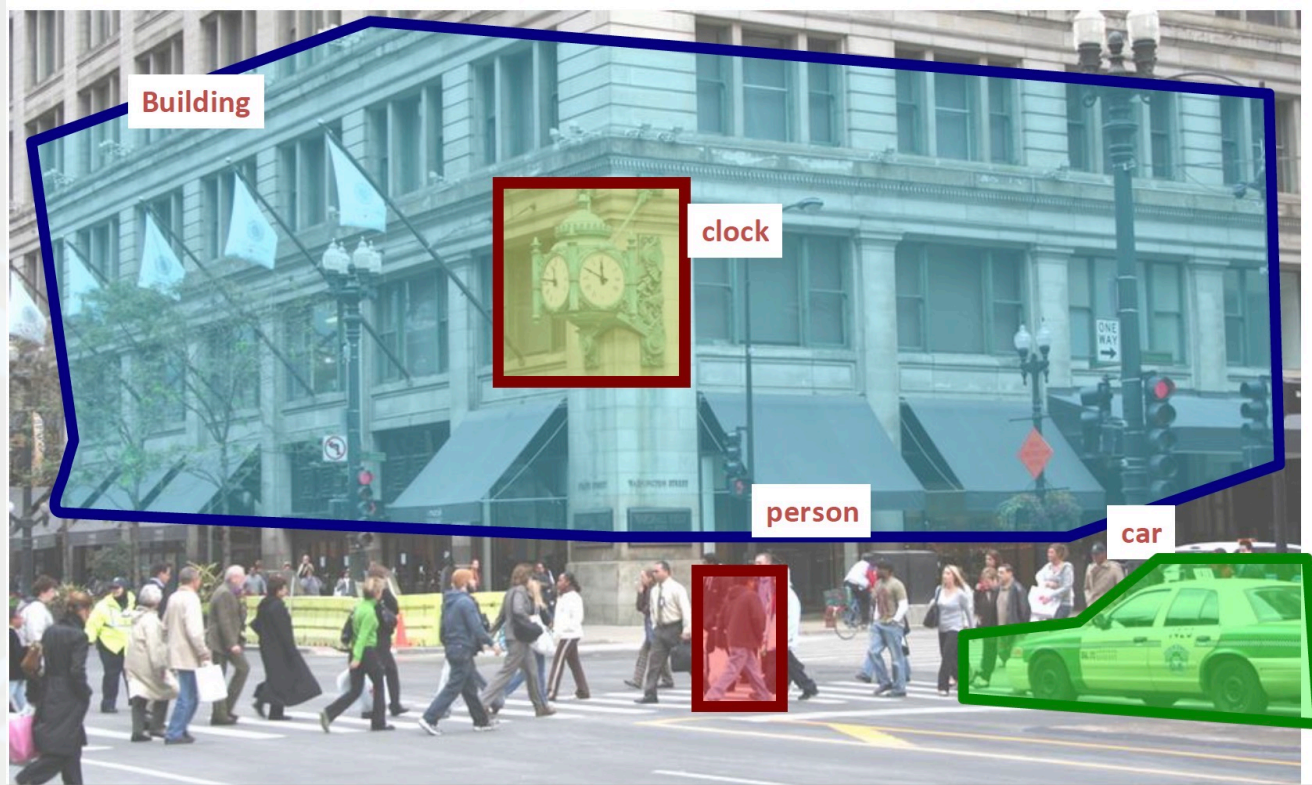
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Detection: Does this image contain a car and where?



Detection: Which objects does this image contain and where?



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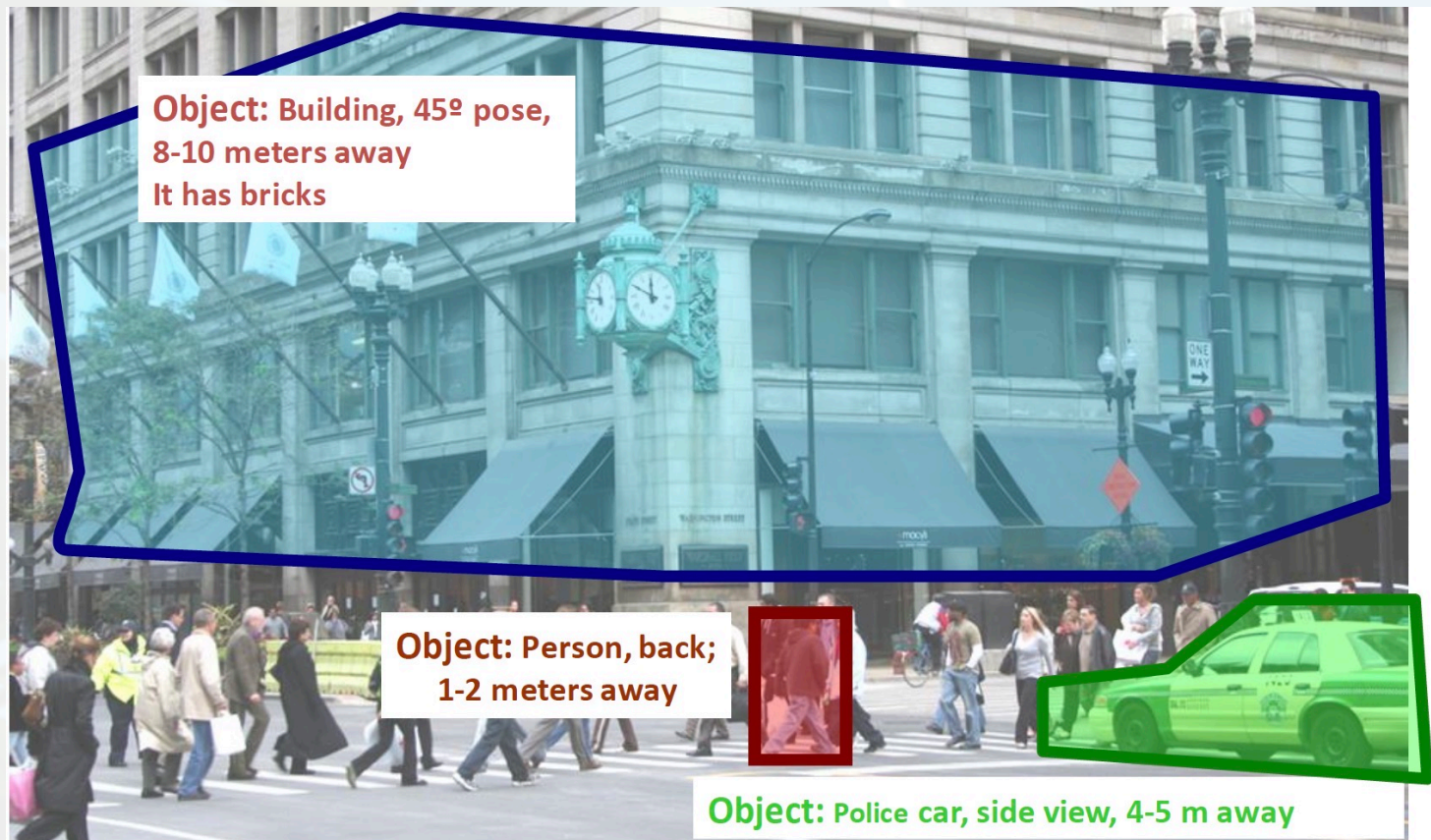
Detection: Accurate localization (segmentation)



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Detection: Estimating object semantic & geometric attributes



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Applications



Computational photography



Security



Assistive driving



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Activity or Event recognition: What are these people doing?



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Applications



Human-Robot Interaction



Sign language

Fig. 2: Ambient intelligence for hospitals.



Ambient Intelligence



Sport analysis



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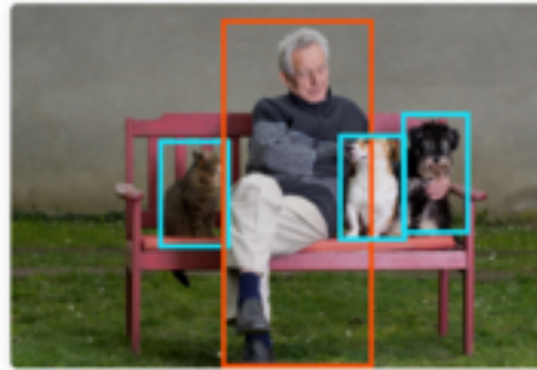
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Classification vs Detection vs Segmentation

PERSON, CAT, DOG



(A) Classification



(B) Detection



(C) Segmentation

output: class label

class label + box

per-pixel label



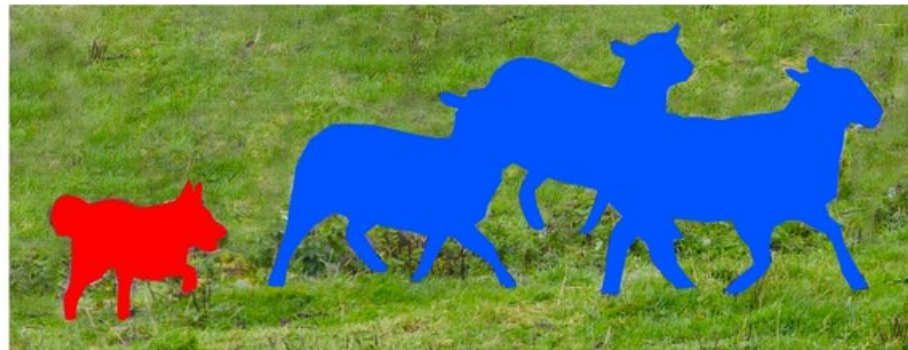
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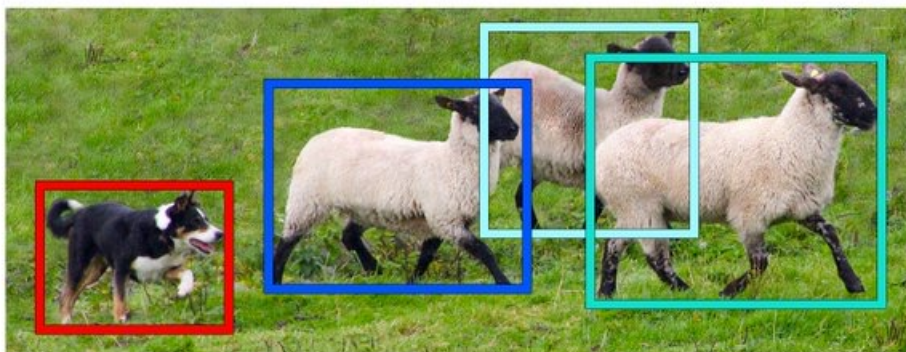
Semantic vs Instance Segmentation



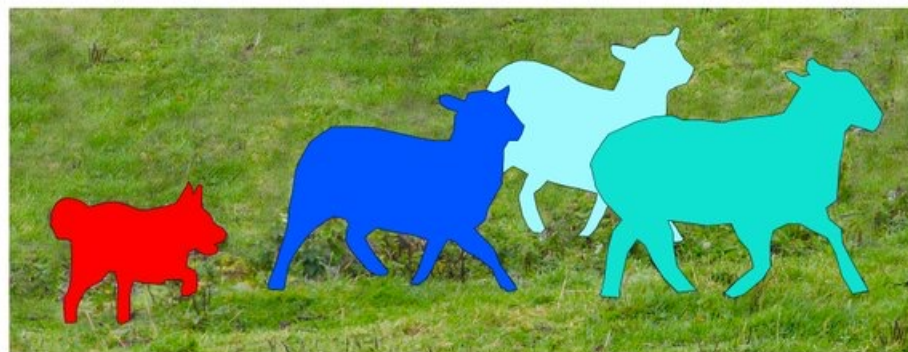
Image Recognition



Semantic Segmentation



Object Detection



Instance Segmentation

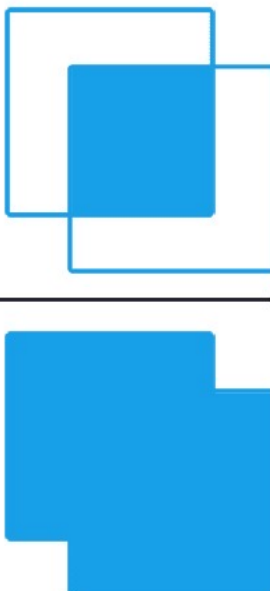


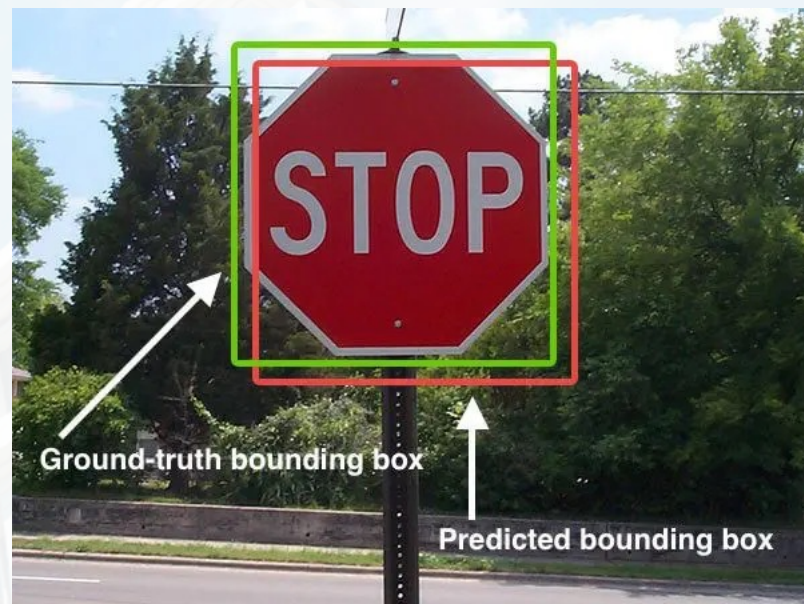
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Object Detection: Evaluation

IoU (Intersection over Union)

$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}}$$




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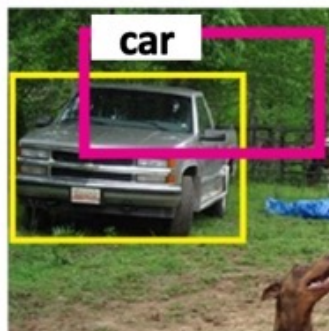
Object Detection: Evaluation



correct
(true positive)



missed
(false negative)



mislocalized
(false positive)

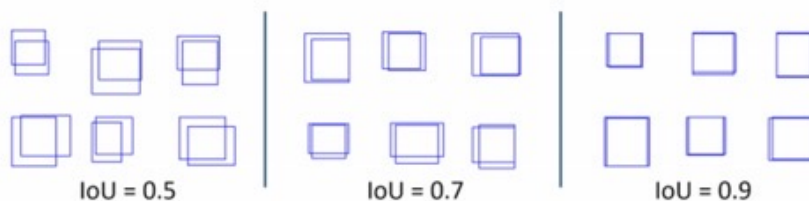


misrecognized
(false positive)



uplicated
(false positive)

define by an IoU threshold



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Object Detection: Evaluation

$$\text{Precision} = \frac{\text{Relevant retrieved instances}}{\text{All retrieved instances}}$$

$$\text{Recall} = \frac{\text{Relevant retrieved instances}}{\text{All relevant instances}}$$

E.g. Consider a computer program for recognizing dogs (the relevant element) in a digital photograph. A picture contains 10 cats and 12 dogs, the program identifies 8 dogs.

Of 8 elements identified as dogs, only 5 actually are dogs (true positives), while the other 3 are cats (false positives). 7 dogs were missed (false negatives), and 7 cats were correctly excluded (true negatives).

The program's precision is then 5/8 (true positives / selected elements) while its recall is 5/12 (true positives / relevant elements).

Figure adapted from: https://en.wikipedia.org/wiki/Precision_and_recall



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Object Detection: Evaluation

- **Precision:** What % of detected objects are correct?

$$\text{Precision} = \frac{\text{true positive}}{\text{true positive} + \text{false positive}}$$

- **Recall:** What % of objects are detected?

$$\text{Recall} = \frac{\text{true positive}}{\text{true positive} + \text{false negative}}$$

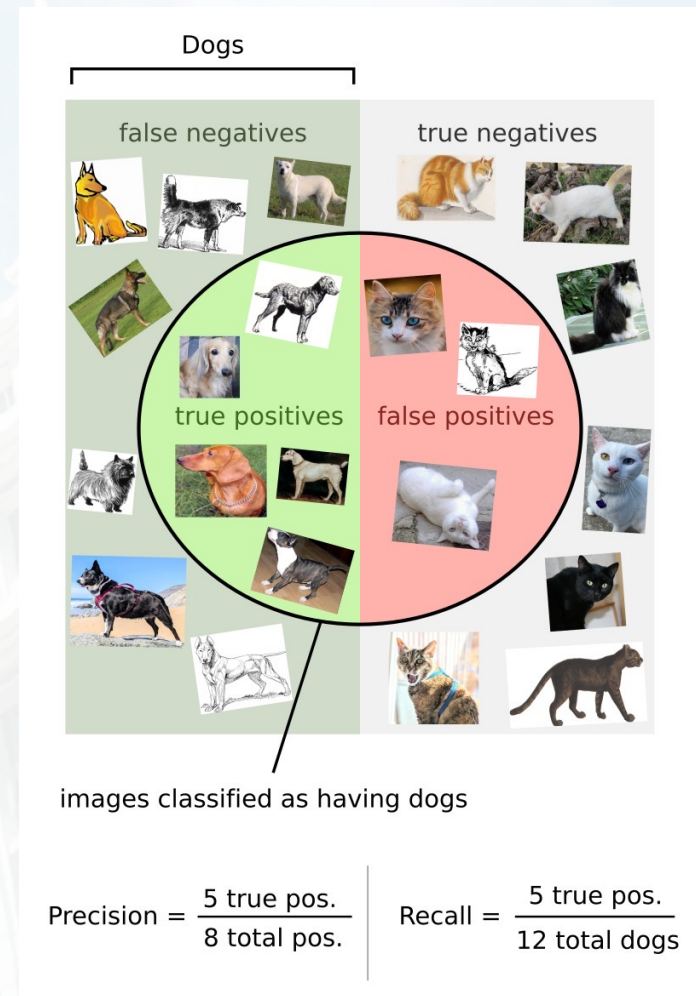


Figure adapted from: https://en.wikipedia.org/wiki/Precision_and_recall



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Object Detection: Evaluation

- **Precision:** What % of detected objects are correct?

$$\text{Precision} = \frac{\text{true positive}}{\text{true positive} + \text{false positive}}$$

- **Recall:** What % of objects are detected?

$$\text{Recall} = \frac{\text{true positive}}{\text{true positive} + \text{false negative}}$$

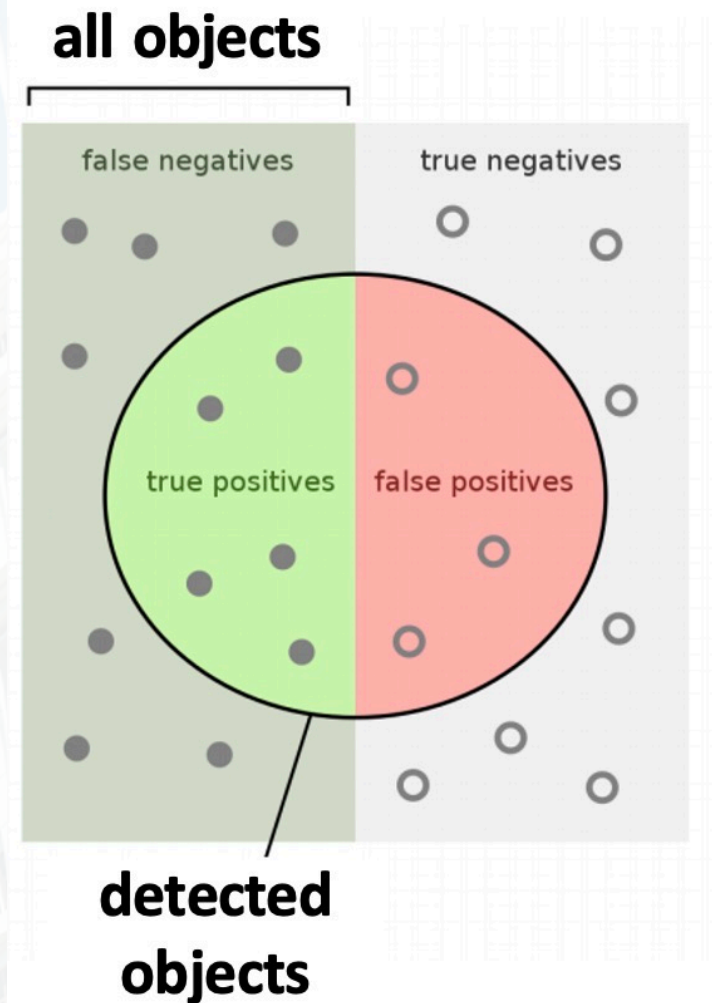


Figure adapted from: https://en.wikipedia.org/wiki/Precision_and_recall

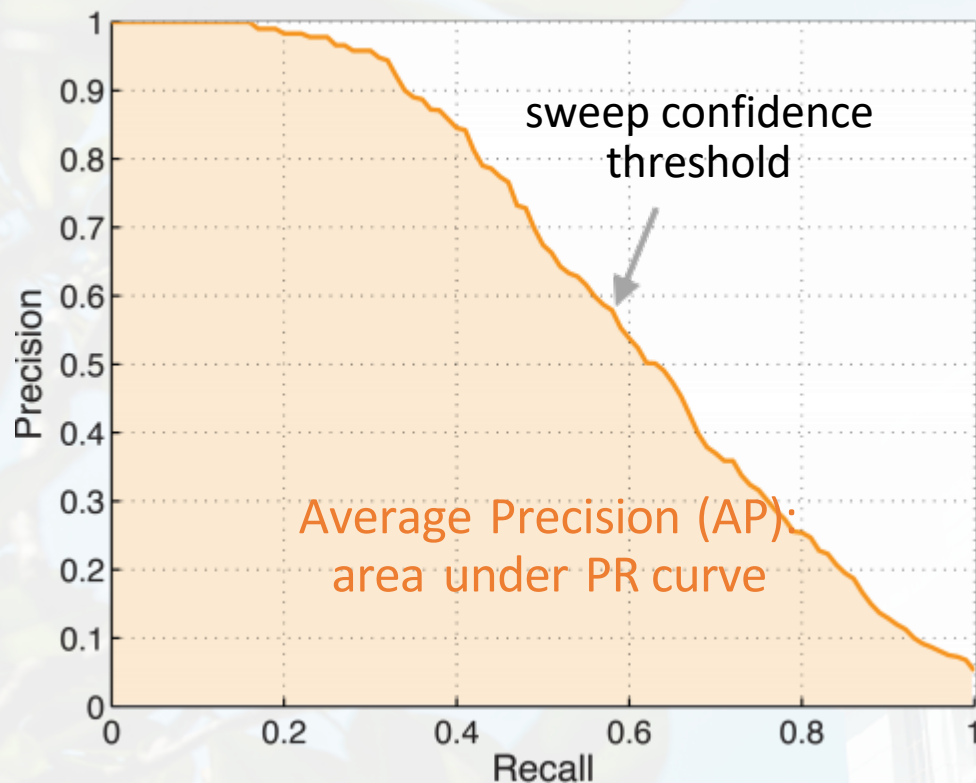


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Object Detection: Evaluation

- Precision-Recall Tradeoff



threshold = **0.4**: Recall ↑, Precision ↓



threshold = **0.7**: Recall ↓, Precision ↑

Life is a precision-recall tradeoff.



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- Introduction to Object Recognition
 - Visual recognition Tasks
 - **Challenges**
- Bias-Variance Trade-off
- A Simple Object Recognition Pipeline



Visual Recognition

- Design algorithms that have the capability to:
 - Classify images or videos
 - Detect and localize objects
 - Estimate semantic and geometrical attributes
 - Classify human activities and events

Why is this challenging?



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How many object categories are there?

~10,000 to 30,000



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Challenges: Viewpoint variation



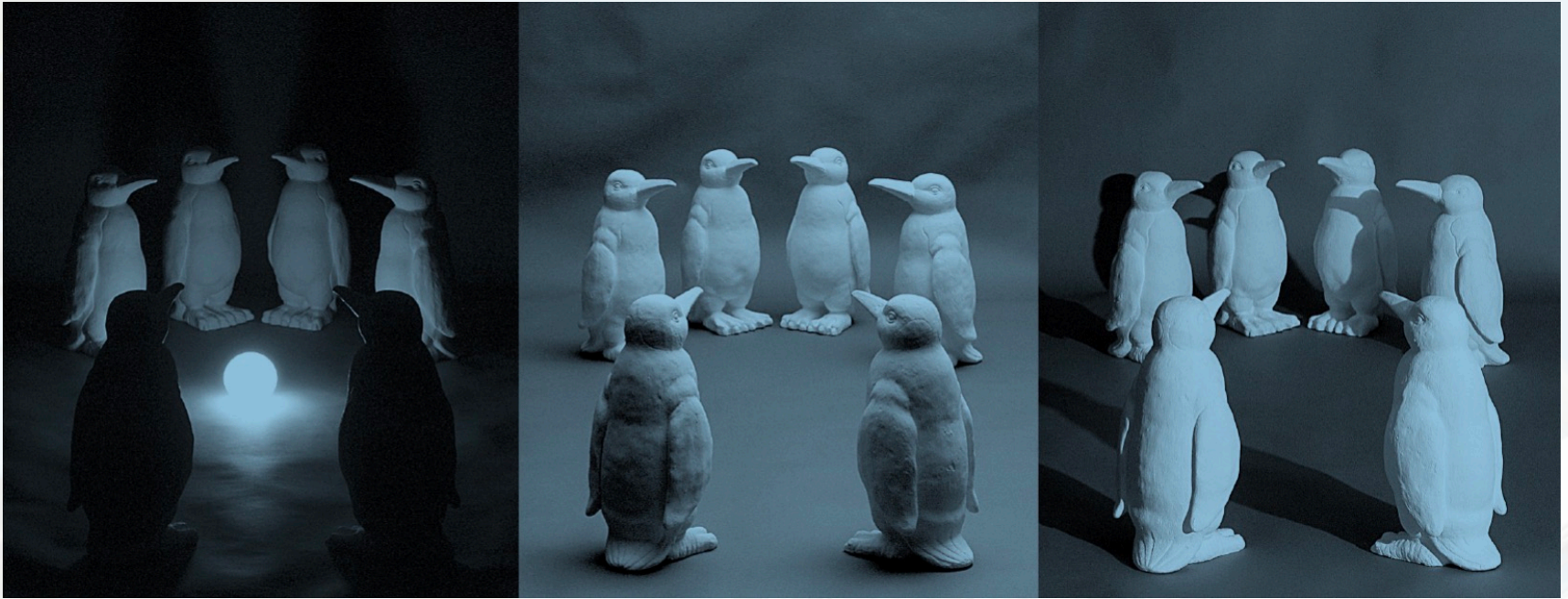
Michelangelo 1475-1564



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Challenges: Illumination



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Challenges: Scale



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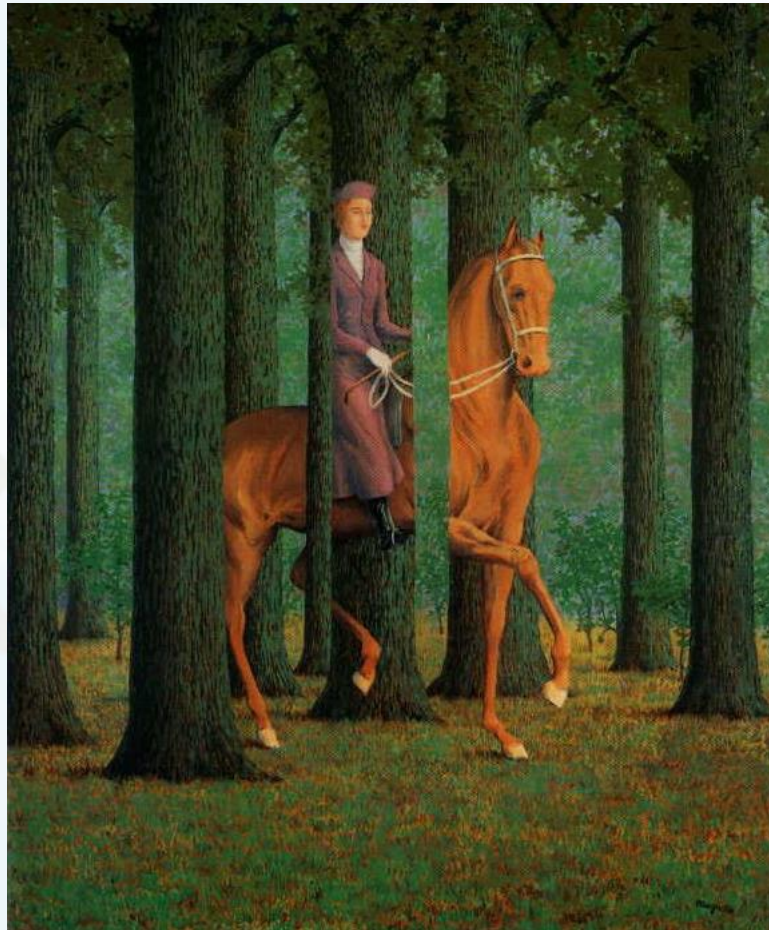
Challenges: Deformation



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Challenges: Occlusion



Magritte, 1957



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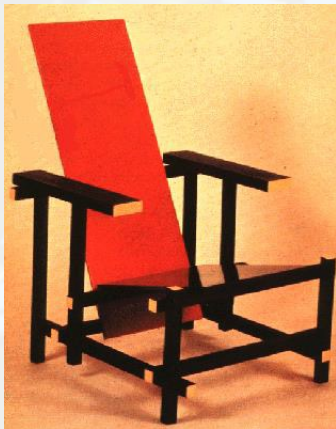
Challenges: Background clutter



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Challenges: Intra-class variation



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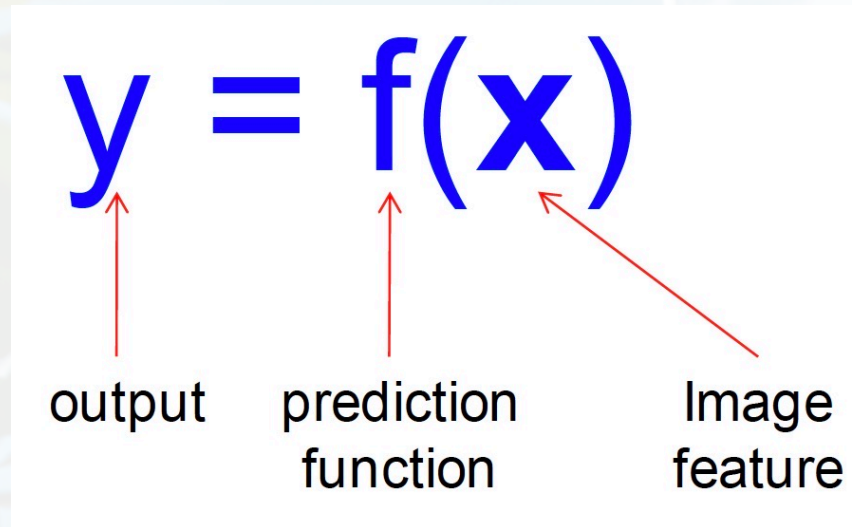
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The machine learning framework

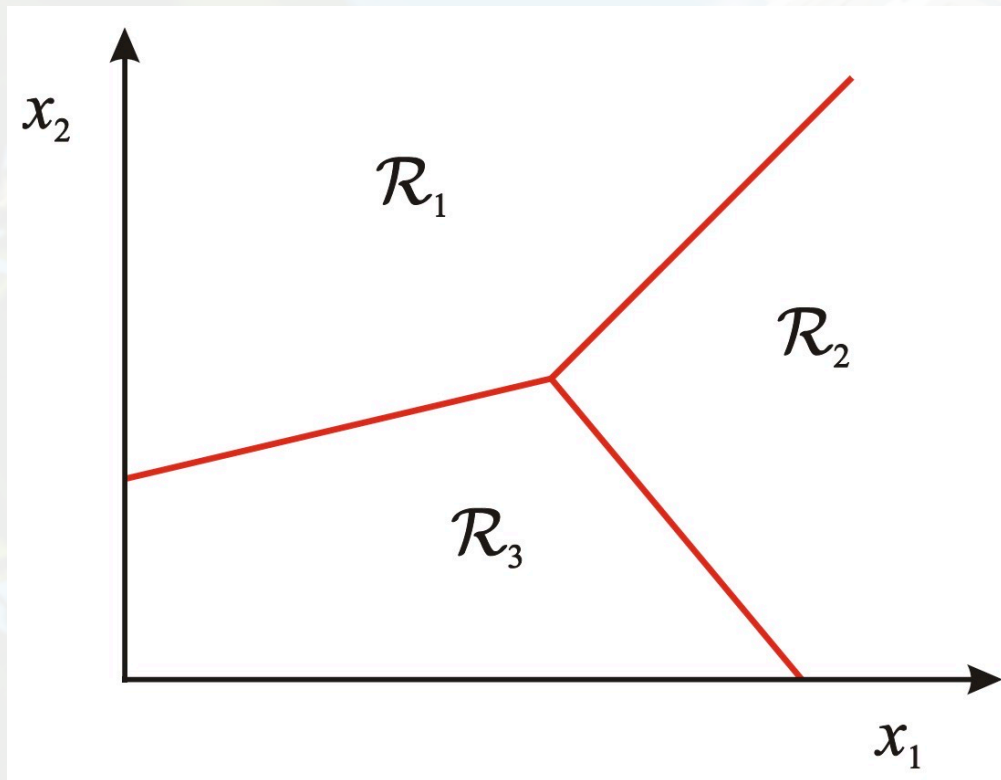


- **Training:** given a *training set* of labeled examples $\{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_N, y_N)\}$, estimate the prediction function f by minimizing the prediction error on the training set
- **Testing:** apply f to a never before seen *test example* $\mathbf{x}$ and output the predicted value $y = f(\mathbf{x})$



Classification

- Assign input vector to one of two or more classes
- Any decision rule divides input space into *decision regions* separated by *decision boundaries*



Many classifiers to choose from

- K-Nearest Neighbor
- SVM
- Neural Networks
- Naïve Bayes
- Bayesian Network
- Logistic Regression
- Randomized Forests
- Boosted Decision Trees
- RBMs
- Etc.

Which is the best one?



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Generalization



Training set (labels known)



Test set (labels unknown)

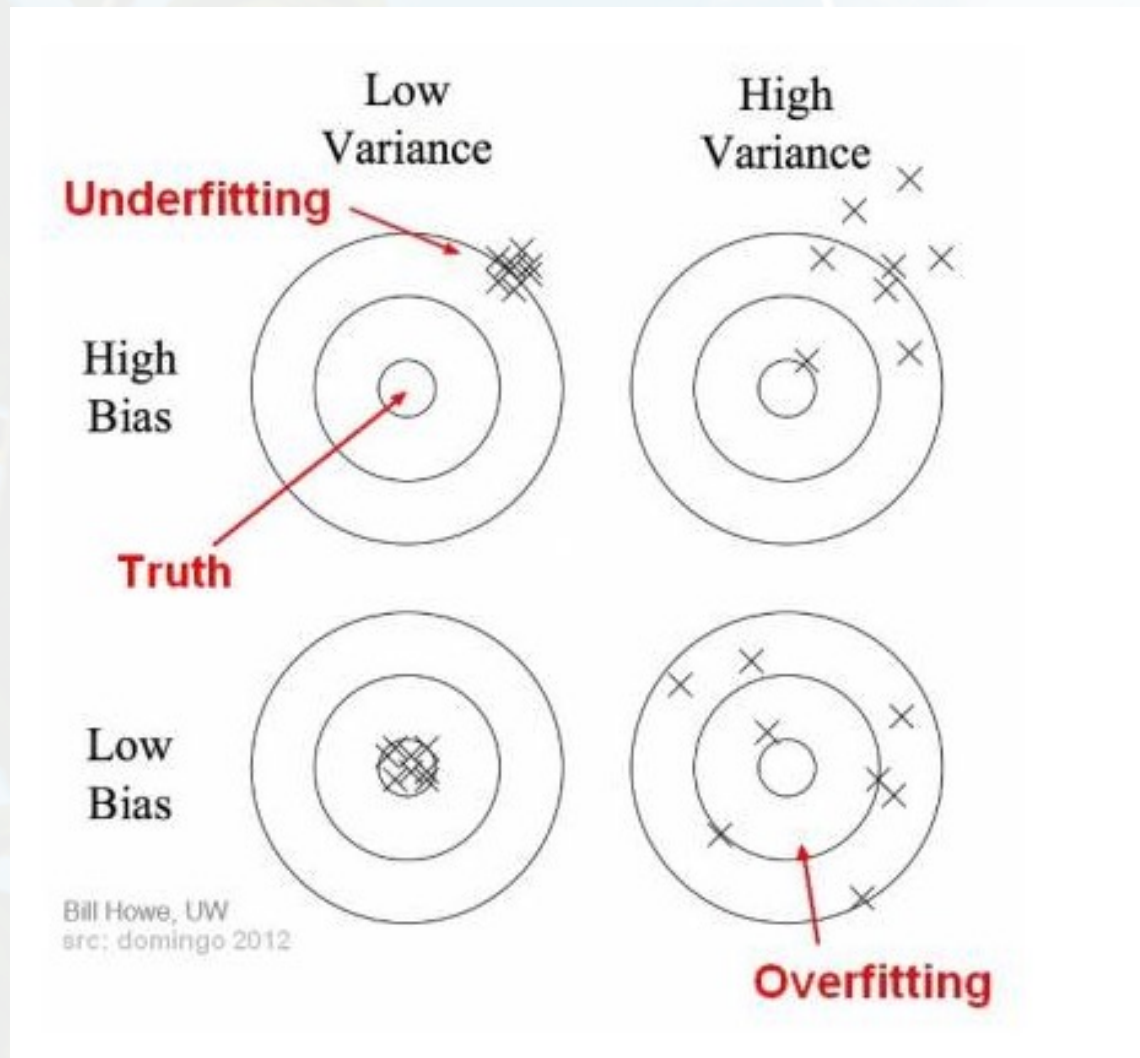
- How well does a learned model generalize from the data it was trained on to a new test set?



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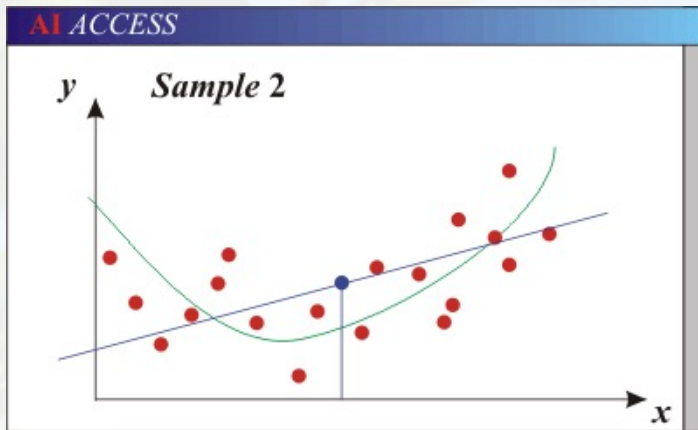
Bias and Variance



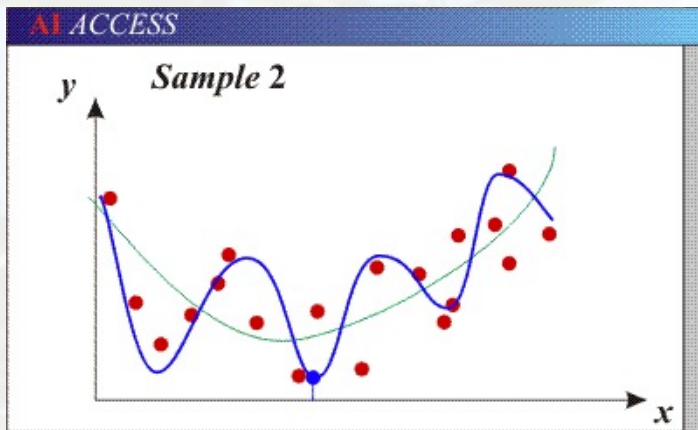
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Bias and Variance



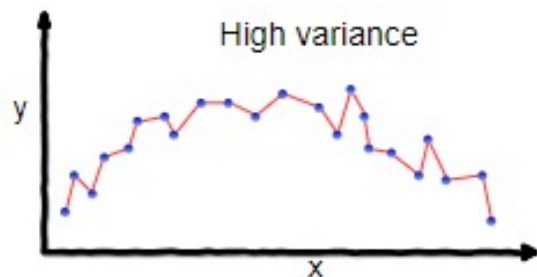
- Models with too few parameters are inaccurate because of a large bias (not enough flexibility).



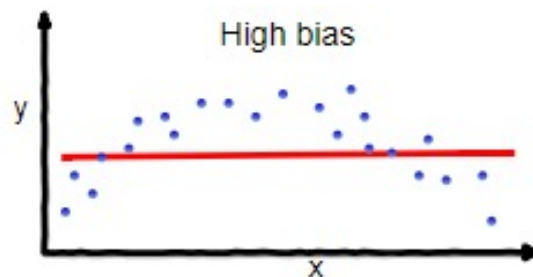
- Models with too many parameters are inaccurate because of a large variance (too much sensitivity to the sample).



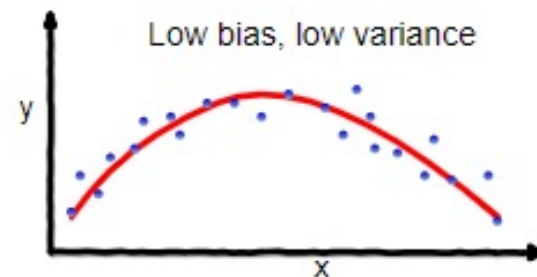
Bias and Variance



overfitting



underfitting



Good balance



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Bias vs Variance

Components of generalization error

–**Bias**: how much the average model over all training sets differ from the true model?

- Error due to inaccurate assumptions/simplifications made by the model

–**Variance**: how much models estimated from different training sets differ from each other



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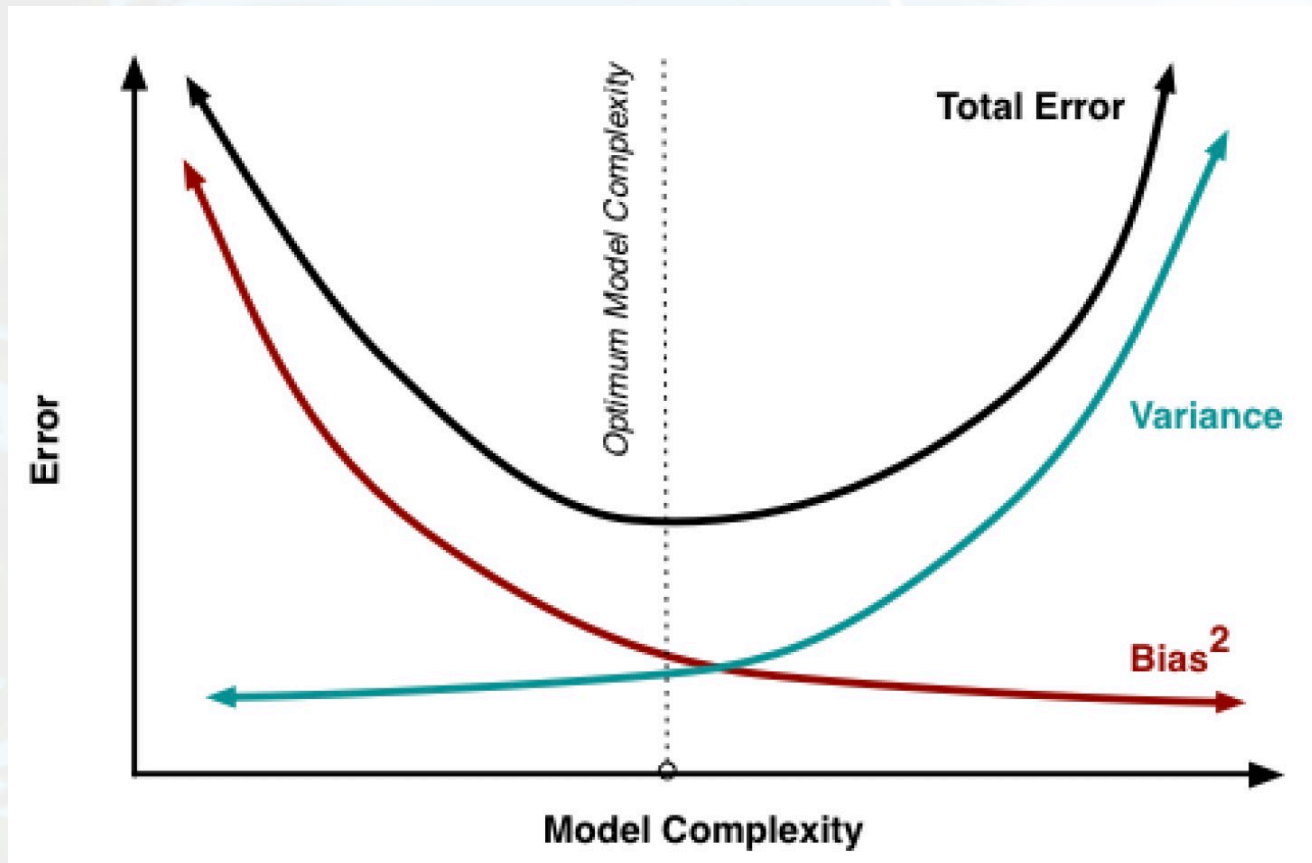
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Bias vs Variance

- **Underfitting:** model is too “simple” to represent all the relevant class characteristics
 - High bias and low variance
 - High training error and high test error
- **Overfitting:** model is too “complex” and fits irrelevant characteristics (noise) in the data
 - Low bias and high variance
 - Low training error and high test error



Bias vs Variance Tradeoff



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Remember...

- No classifier is inherently better than any other: you need to make assumptions to generalize
- Three kinds of error
 - Inherent: irreducible error
 - Bias: due to over-simplifications
 - Variance: due to inability to perfectly estimate parameters from limited data
- Derivation: see in class



How to reduce variance?

- Choose a simpler classifier
- Regularize the parameters
- Get more training data



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Last remarks about applying machine learning methods to object recognition

- There are machine learning algorithms to choose from
- Know your data:
 - How much supervision do you have?
 - How many training examples can you afford?
 - How noisy?
- Know your goal (i.e. task):
 - Affects your choices of representation
 - Affects your choices of learning algorithms
 - Affects your choices of evaluation metrics
- Understand the math behind each machine learning algorithm under consideration!



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Object recognition: an image classification framework

- Apply a prediction function to a feature representation of the image to get the desired output:

$f(\text{apple}) = \text{"apple"}$

$f(\text{tomato}) = \text{"tomato"}$

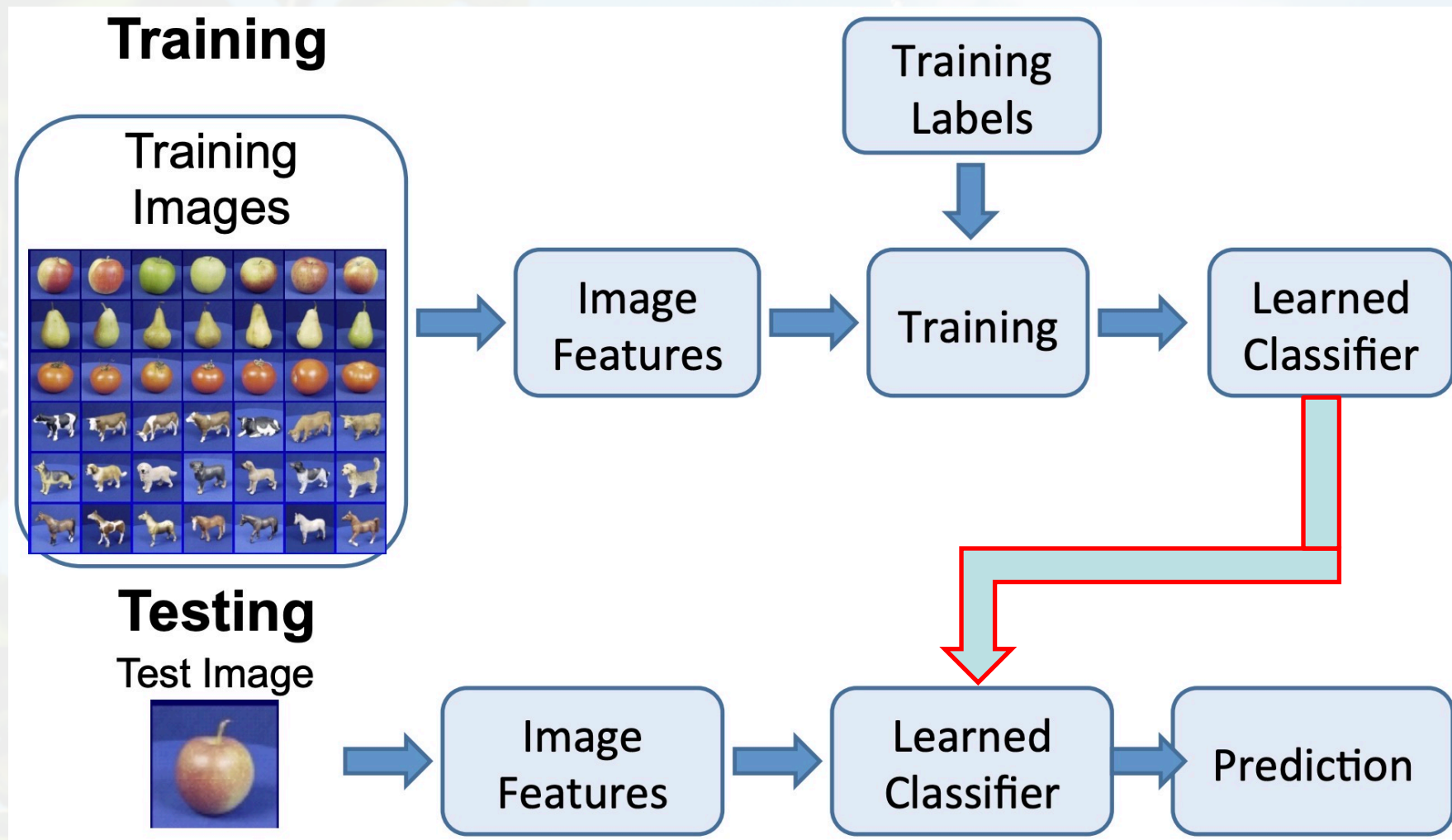
$f(\text{cow}) = \text{"cow"}$



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A simple pipeline

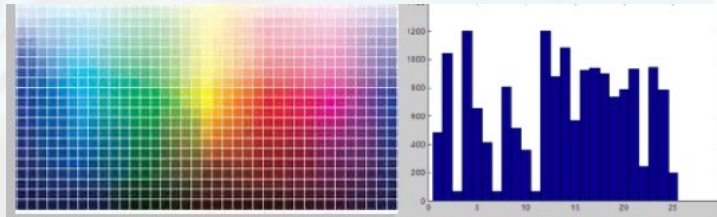


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Image features

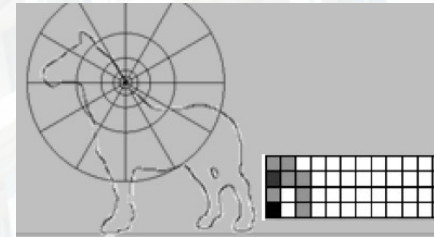
- Color



- Global shape: PCA space



- Local shape: shape context



- Texture: filter banks

